Label Quality

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Label Quality before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Label Quality trace and record the first step you can explain without help.

Use W2: Interpret **agreement** for Label definitions and dis**agreement**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

Check the proposed Label Quality explanation against the supplied facts before discussing it.

Use the worked case for Label definitions and dis**agreement**. Proposed explanation: "**Agreement** is accuracy". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply the process independently

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Label Quality target to a trace and a justified conclusion.

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Label Quality case and show the deciding evidence.

Two **annotator**s disagree on three of ten leaves, especially mild spotting.

Improve label quality.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better **process**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 6 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	

Test record	My evidence
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Label Quality adds to Responsible Plant Classifier and keep the evidence.

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Use the boundary and review rules

> Use the boundary and review rules. Apply the supplied Label Quality facts and keep your reasoning.

Under rule v1, label a wheel touching the boundary, an object entirely outside a fully visible lane, and a partly hidden lane. Explain the **evidence** for each.

My reasoning and evidence

W2 Interpret agreement

> Interpret agreement. Apply the supplied Label Quality facts and keep your reasoning.

Explain the **output** 2 4 for A through D. State the fraction and percent, whether review is included, and why this is not model accuracy.

My reasoning and evidence

W3 Write a decision log

> Write a decision log. Apply the supplied Label Quality facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Label Quality", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For B and C, record original labels, visible **evidence**, rule version and reviewed decision. Explain why the original labels should remain available.

My reasoning and evidence

W4 Apply the process independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply the process independently. Apply the supplied Label Quality facts and keep your reasoning.

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Draft and test a label rule

> Draft and test a label rule. Apply the supplied Label Quality facts and keep your reasoning.

Write a short v1 labeling instruction covering blocked, clear, the touching boundary and review. Add one boundary example. Explain why three people voting clear cannot settle a hidden lane.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Label Quality

> Write a label rule that resolves the visible boundary case. Explain the labelled step before checking the changed condition.

Write a **label rule** that resolves the visible boundary case.

Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review. X has one visible spot; Y has none; Z is half hidden.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/6

Label Quality

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Apply visible rule

X is spotted; Y is healthy under this exercise's definition.



2. Handle uncertainty

Z requires review; hidden tissue cannot be treated as visibly spot-free.



3. Review disagreement

If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

CHANGE ONE THING

Z receives a new fully visible image with no brown spots.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a **label rule** that resolves the visible boundary case.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.

1. Apply visible rule

2. Review dis**agreement**

3. Handle uncertainty

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Review dis**agreement** If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Read the picture in order.

If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Z receives a new fully visible image with no brown spots. Which conclusion is supported by the supplied facts?

1. Label the new **observation** healthy under the toy rule.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Label the new **observation** healthy under the toy rule. The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Repair a missing building block

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

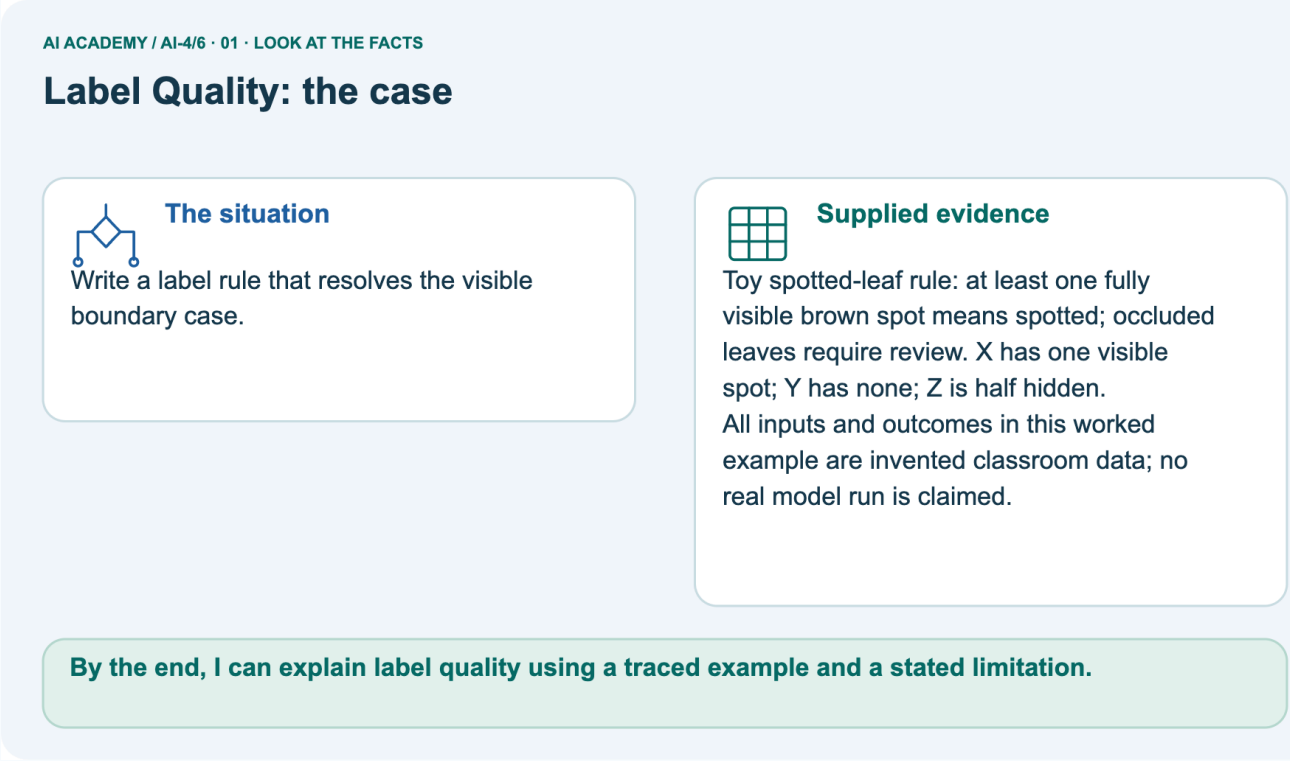
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Label Quality picture from its labels, then record a relevant limit.

01. Label Quality: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

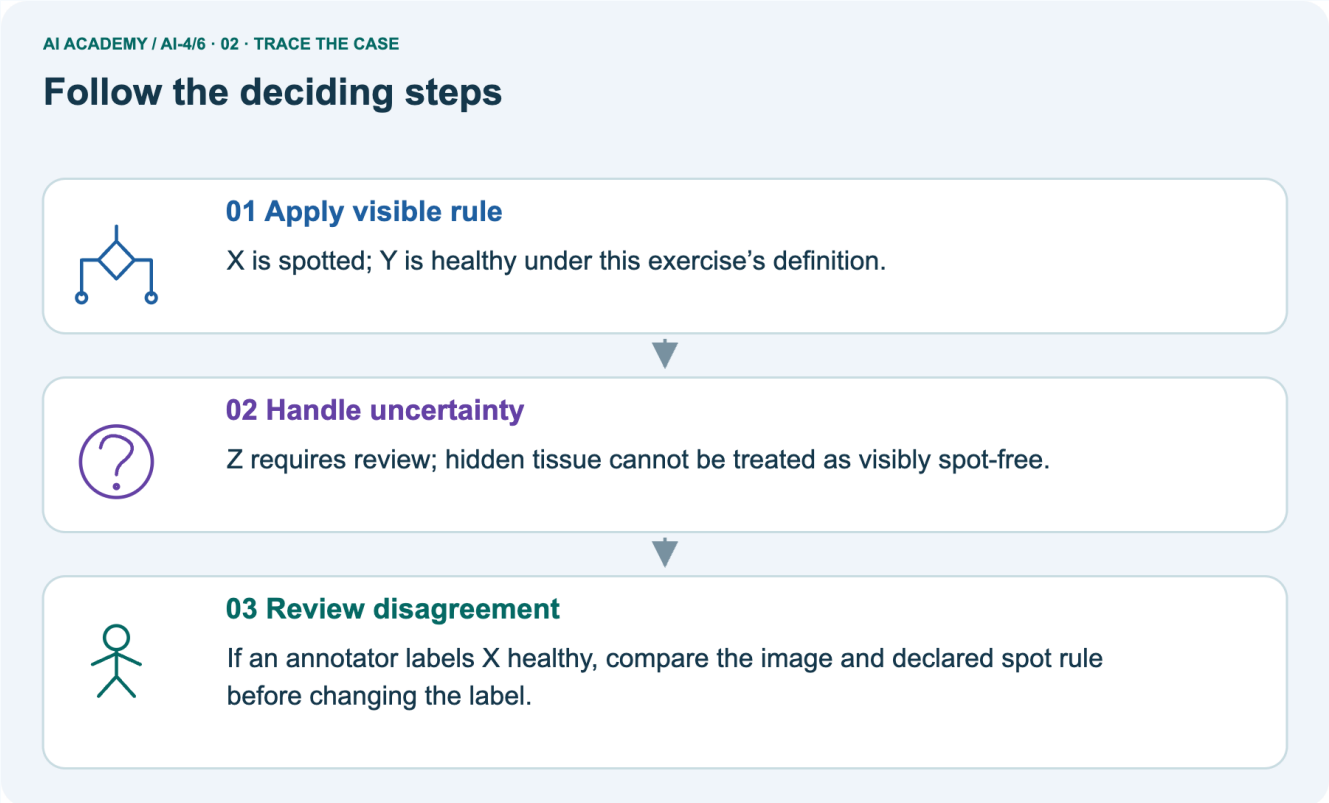
Write a **label rule** that resolves the visible boundary case. Supplied facts: Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review.

X has one visible spot; Y has none; Z is half hidden. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Apply visible rule: X is spotted; Y is healthy under this exercise's definition. → Handle uncertainty: Z requires review; hidden tissue cannot be treated as visibly spot-free. → Review dis**agreement**: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow a disagreement back to evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

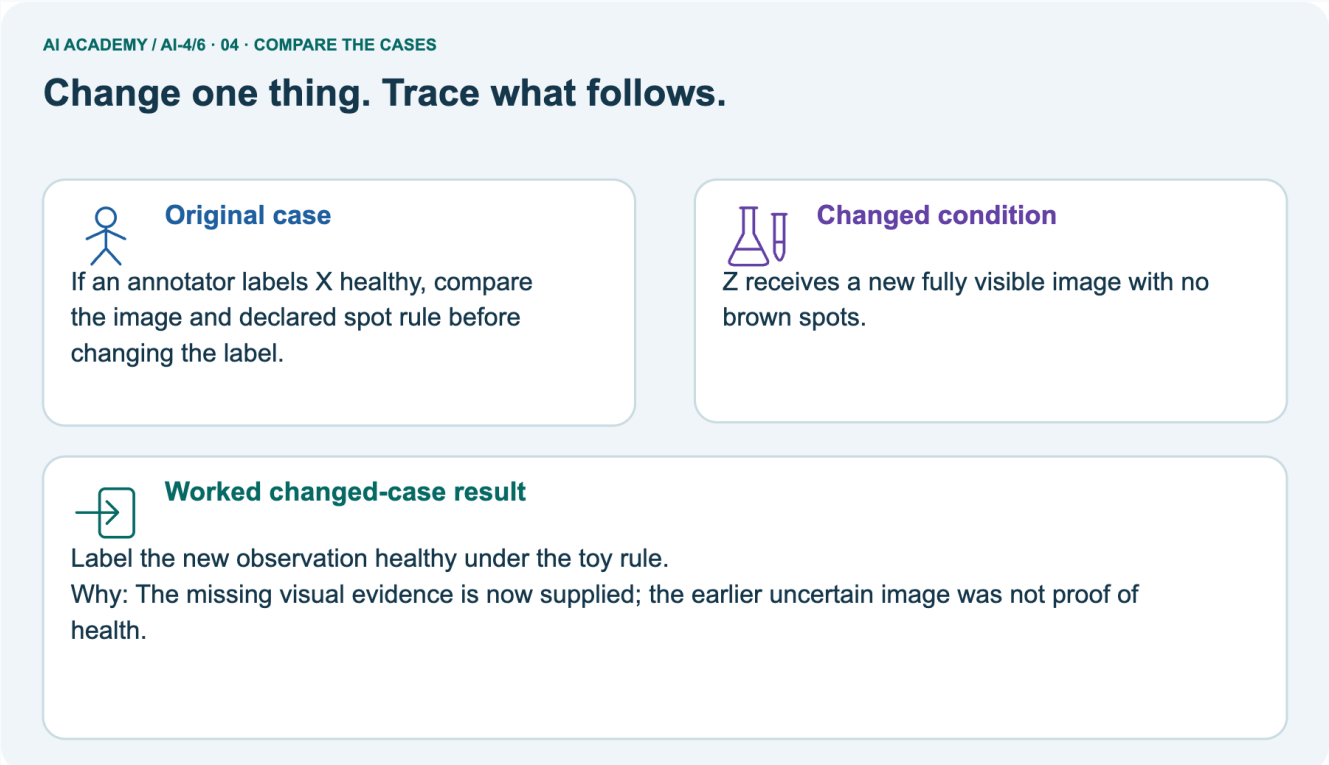
Follow a dis**agreement** back to **evidence**. B has two labels | Read boundary rule | Keep labels and reason. C includes review | Check visible **evidence** | Retain uncertainty if needed.

A and D match | Check rule as well | **Agreement** is not proof

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

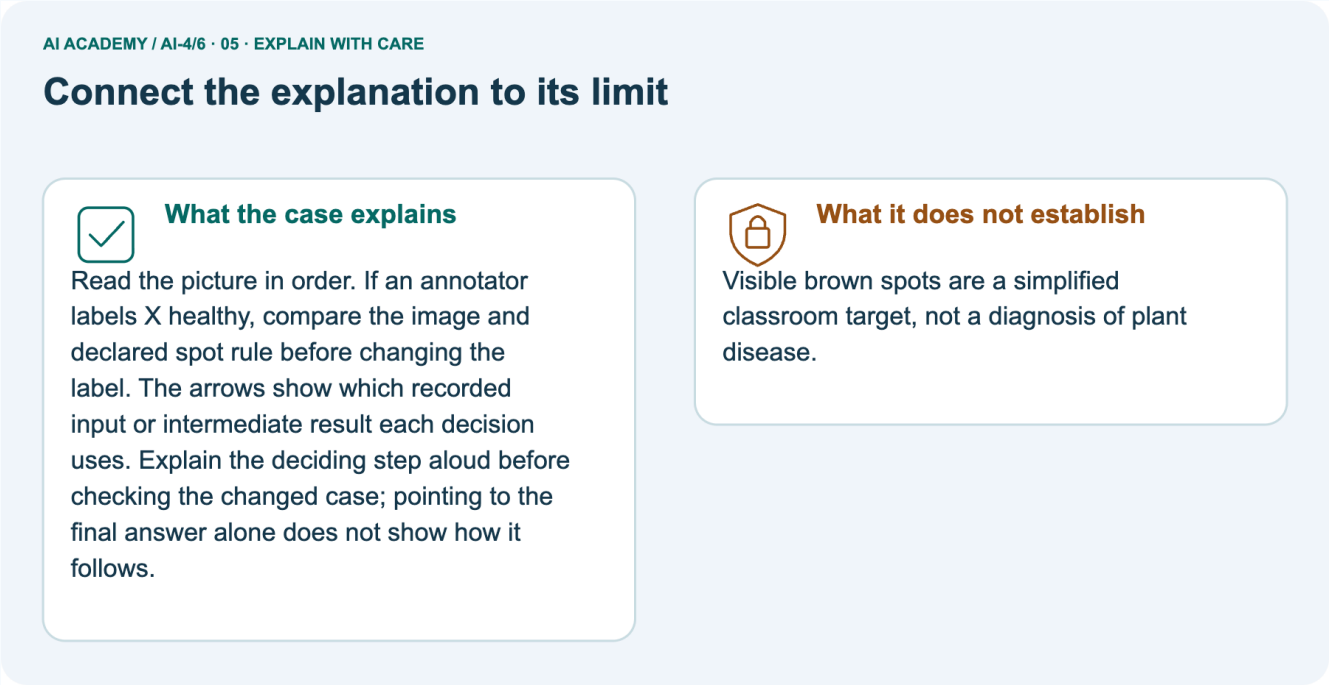
Original case: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Changed condition: Z receives a new fully visible image with no brown spots.

Expected result: Label the new **observation** healthy under the toy rule. Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

> Explain the case, then point to a limit on the claim.

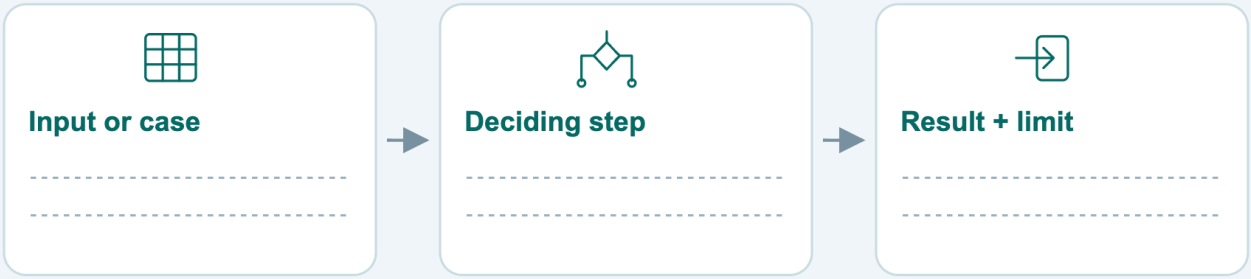
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-4/6 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Why should two annotators not vote away Z's missing evidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.